

3rd PFITS+ Meeting on Hydrodynamics of Circumstellar Disks and Planet Formation

Impact of vertical shear instability on dust evolution

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Fukuhara, Flock, Okuzumi, Tominaga, A&A, 701, A72 (2025)

One of candidates for turbulence-driving mechanisms

e.g., Nelson et al. 2013; Lin & Youdin 2025

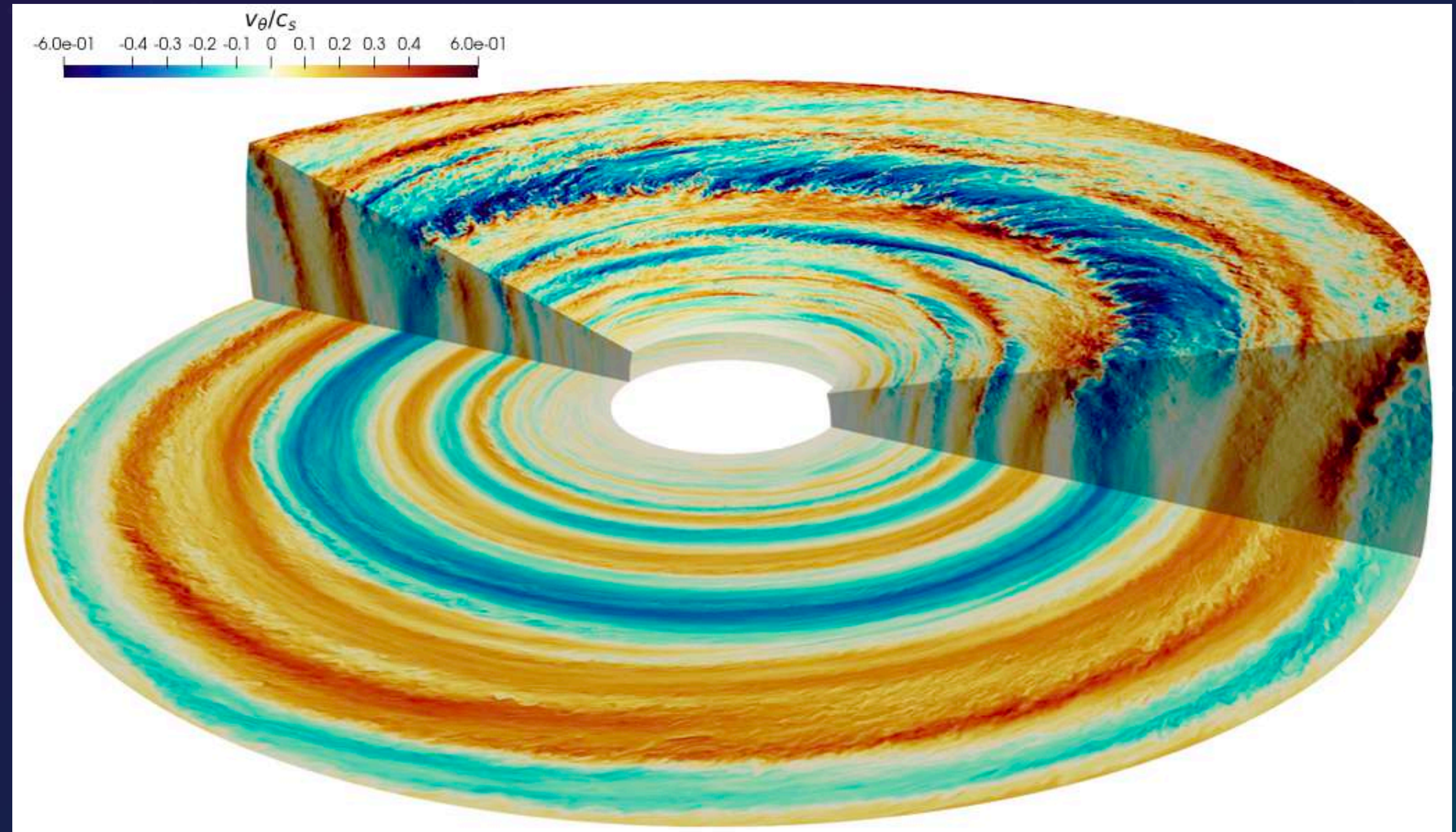
3D hydrodynamical simulation of VSI turbulence

- ❖ VSI turbulence can diffuse dust vertically

e.g., Stoll & Kley 2016; Flock et al. 2020; **Fukuhara** et al. 2025

- ❖ VSI turbulence depends on cooling timescales

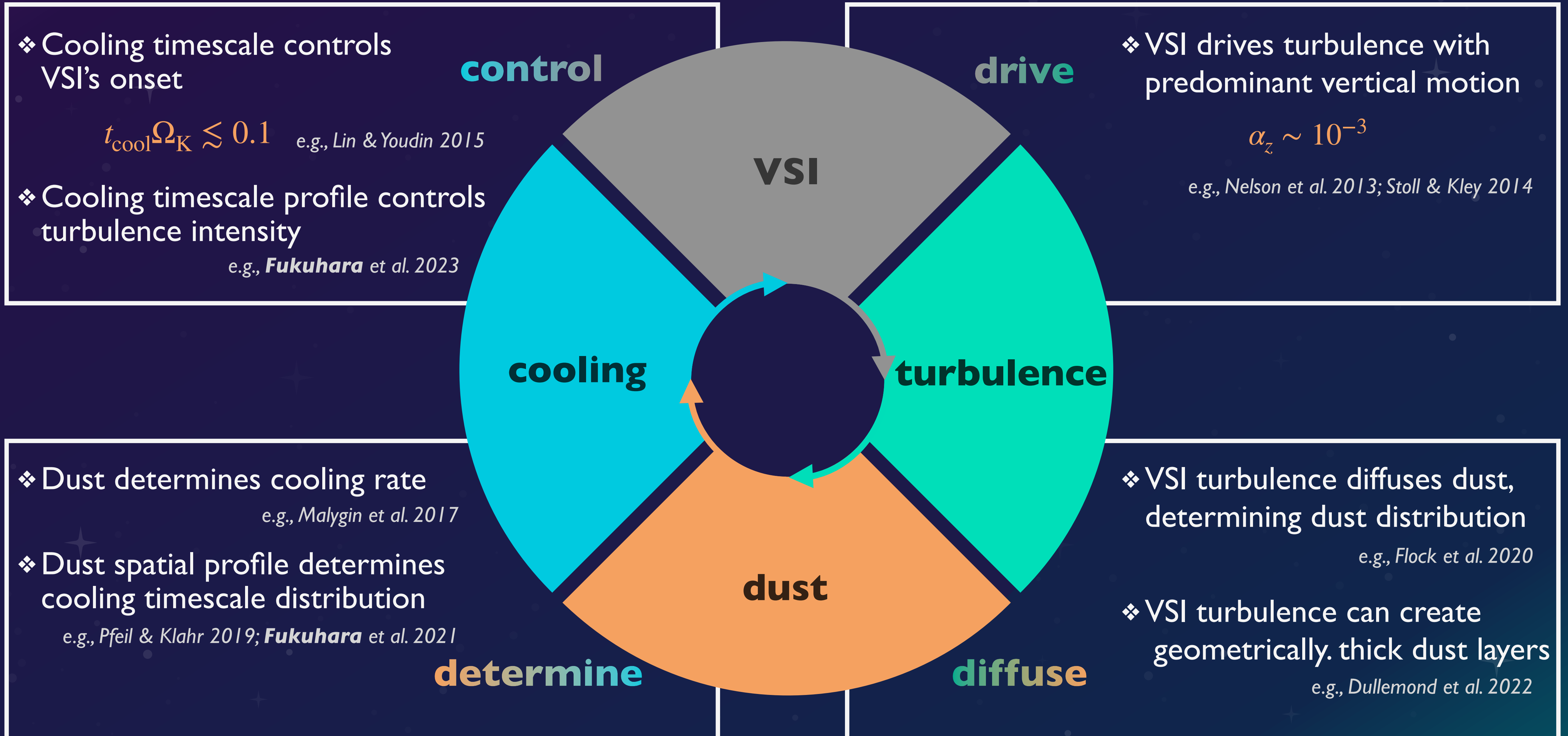
e.g., Lin & Youdin 2015; Pfeil & Klahr 2021; **Fukuhara** et al. 2023



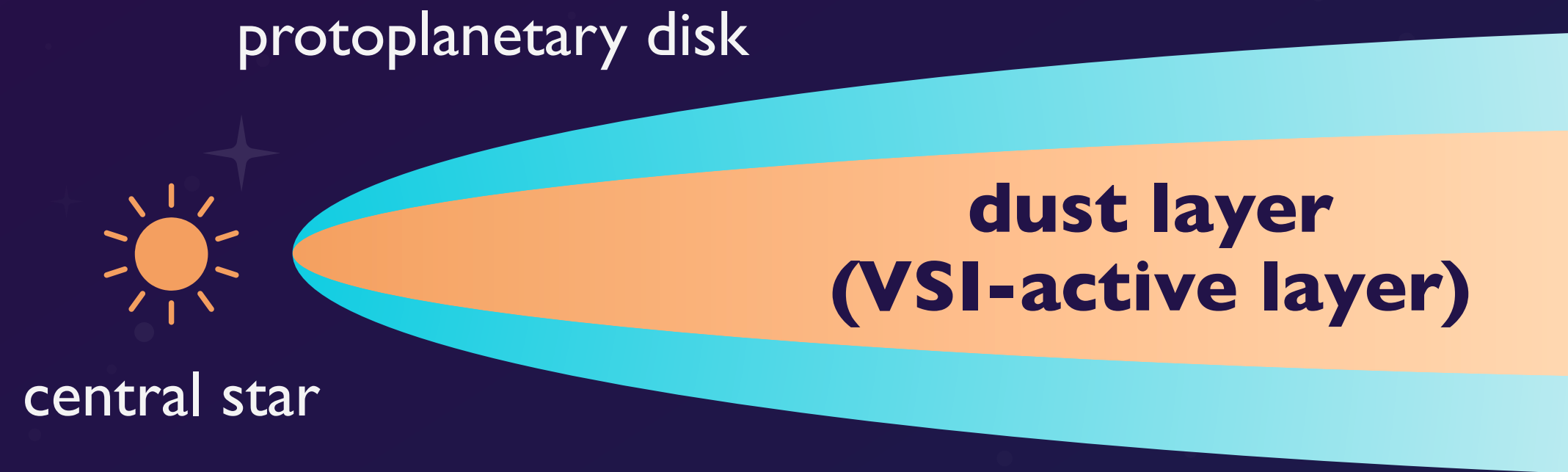
Lesur et al. 2025

Interaction!

Pfeil et al. 2023,2024; Fukuhara & Okuzumi 2024



Hydrodynamical simulations with dynamic dust and cooling timescale



- ❖ global 2.5D axisymmetric disk simulations
 - radial-meridional (r, θ) plane
 - three components for velocities (v_r, v_θ, v_ϕ)

- ❖ code: PLUTO with pressureless dust module

Mignone et al. 2007; Ziampras et al. 2025

- ❖ resolution ≈ 70 cells/ H_{gas}

gas

- ❖ cooling time depends on dust
 - assuming optically thin regime
 - main process is collisional heat transfer

e.g., Malygin et al. 2017

$$t_{\text{cool}} = \frac{1}{v_{\text{th}}} \left(\pi a^2 \cdot \frac{\rho_{\text{dust}}}{m_{\text{dust}}} \right)^{-1} \propto \left(\frac{a}{\rho_{\text{dust}}} \right)$$

- ❖ VSI criterion $t_{\text{cool}} \Omega_K \lesssim 0.1$ *e.g., Lin & Youdin 2015*

drag force



cooling time



dust

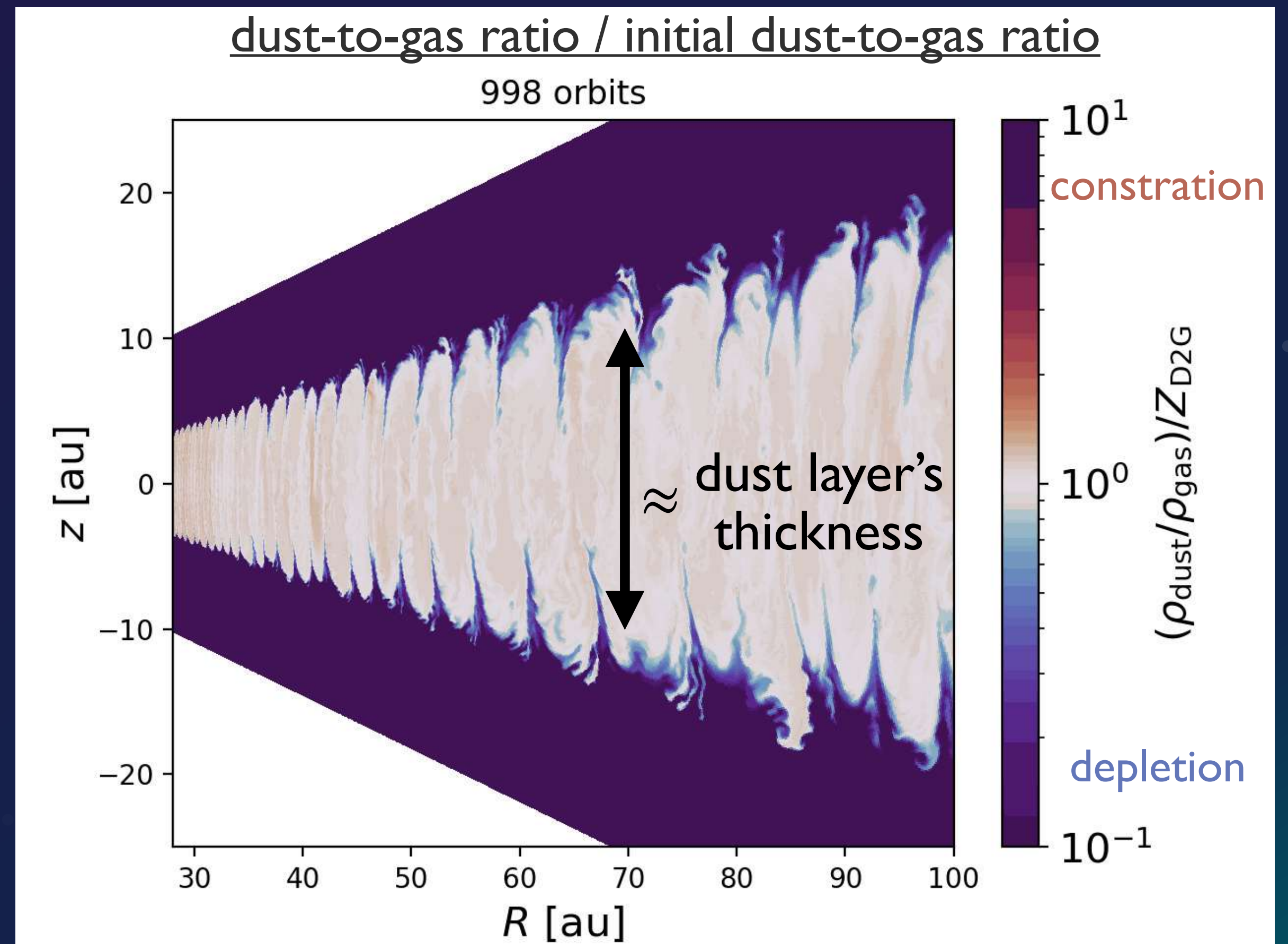
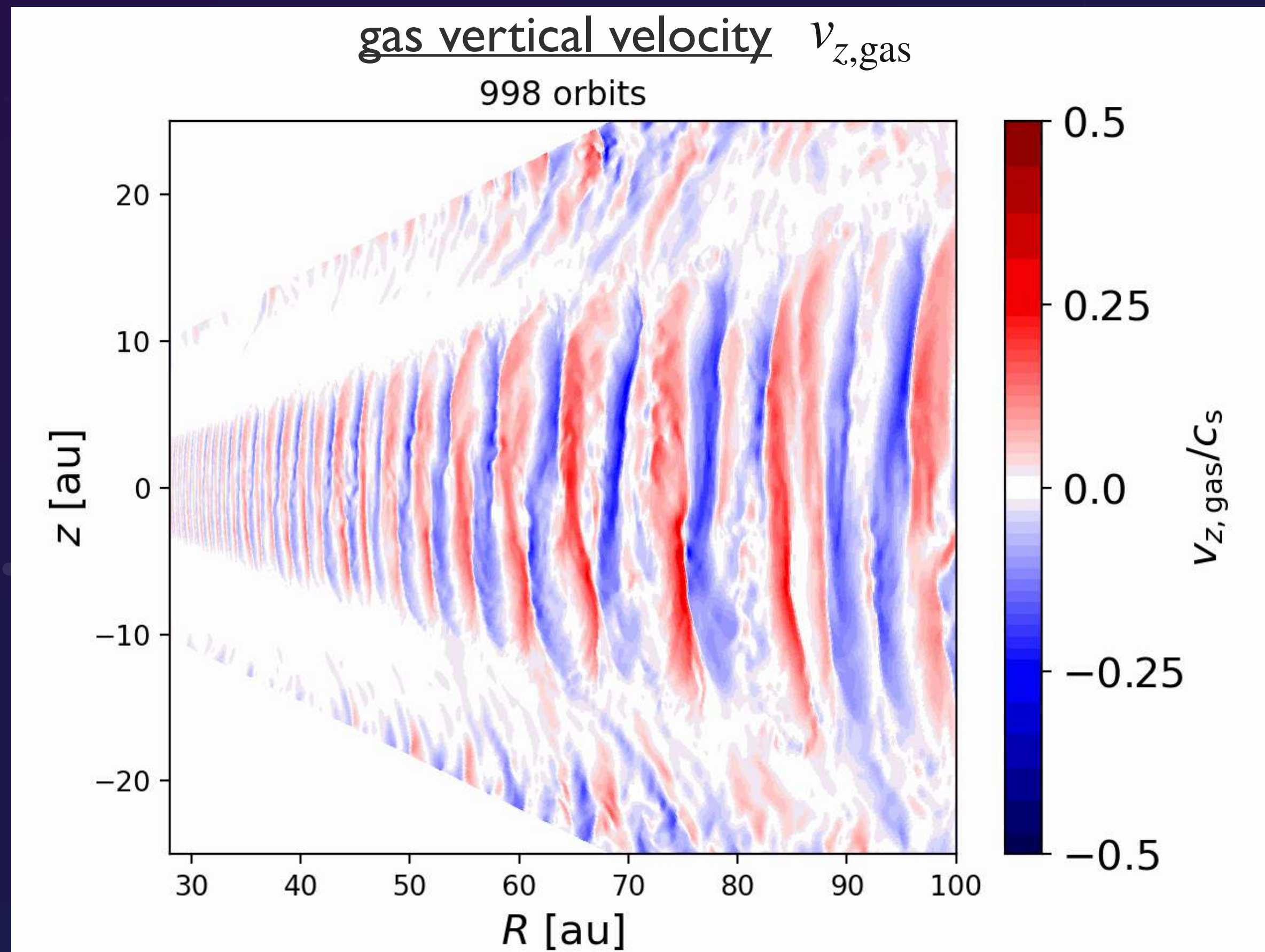
- ❖ single size dust as pressureless fluid
- ❖ motion is determined by **gas drag force**
- ❖ main parameters
 - **dust grain size: a**
 - **global dust-to-gas mass ratio: Z_{D2G}**

Saturated global 2D structure

- ❖ There exists the **equilibrium state** for VSI turbulence and dust profile
- ❖ VSI generates turbulence and sustain dust diffusion, creating a thick dust layer

grain size = $10\ \mu\text{m}$

dust-to-gas ratio = 0.01

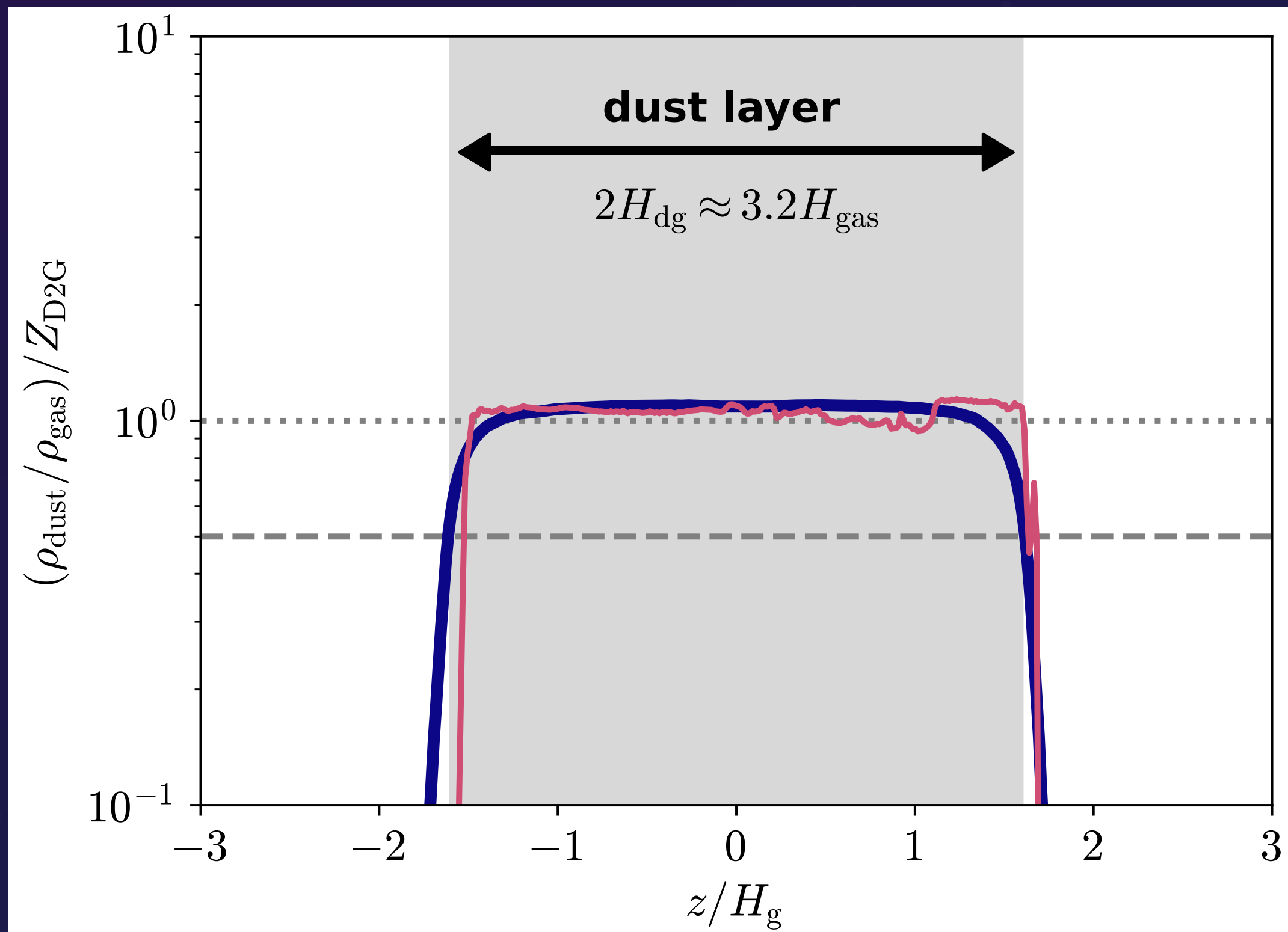


Dust layer's thickness

- ❖ From vertical profile of dust-to-gas density ratio, we define dust layer and its thickness $2H_{\text{dg}}$

$$2H_{\text{dg}} \approx 3.2H_{\text{gas}}$$

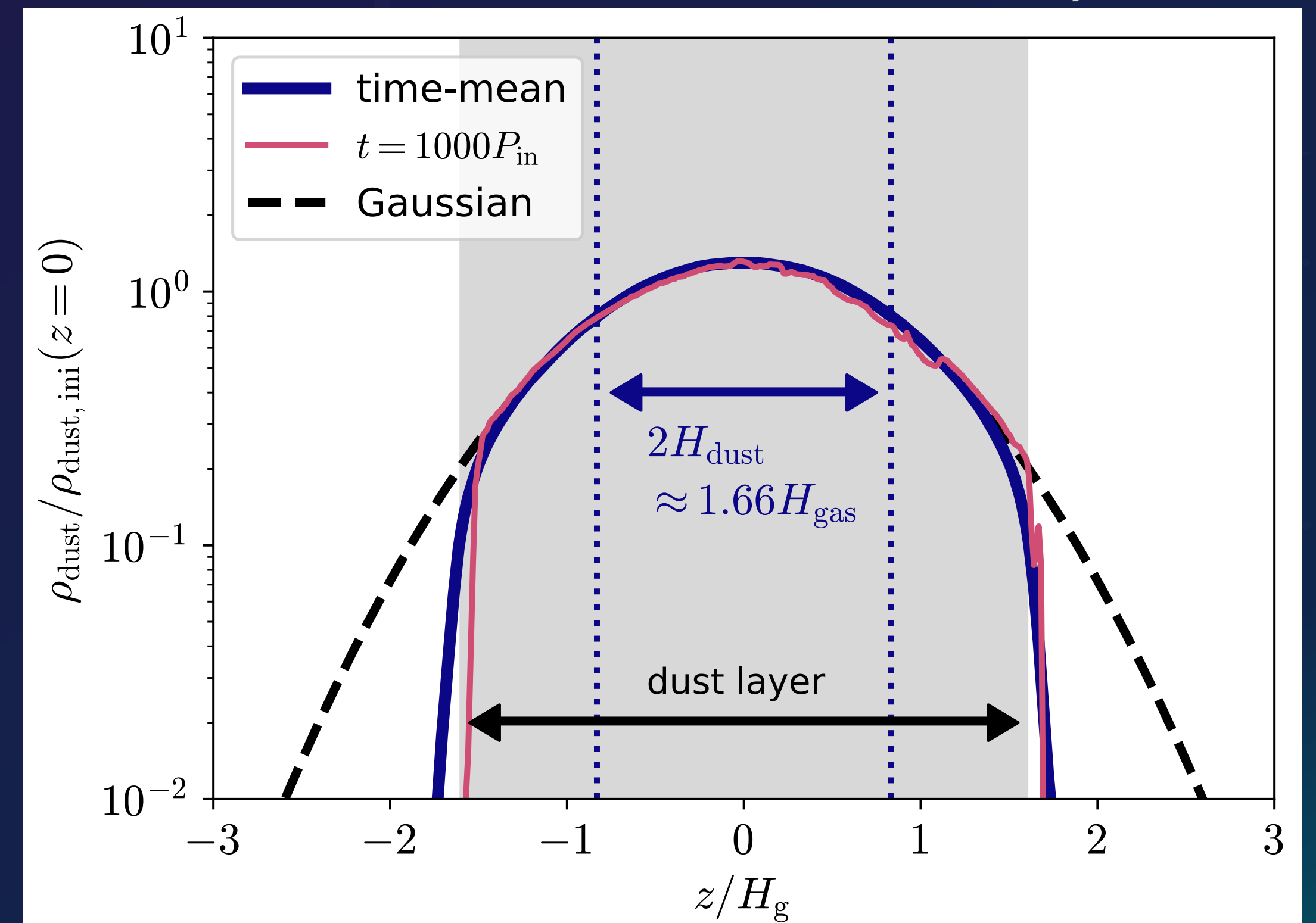
vertical distribution of dust-to-gas density ratio



Dust scale height

- ❖ Dust density follows a Gaussian distribution within dust layer
- ❖ Dust scale height is estimated as $H_{\text{dust}} \approx 0.83H_{\text{gas}}$

vertical distribution of dust density



Quasi-steady state

- ❖ During the early stage, dust layer becomes thinner as dust settles toward the midplane
- ❖ Dust layer's thickness reaches a quasi-steady value of

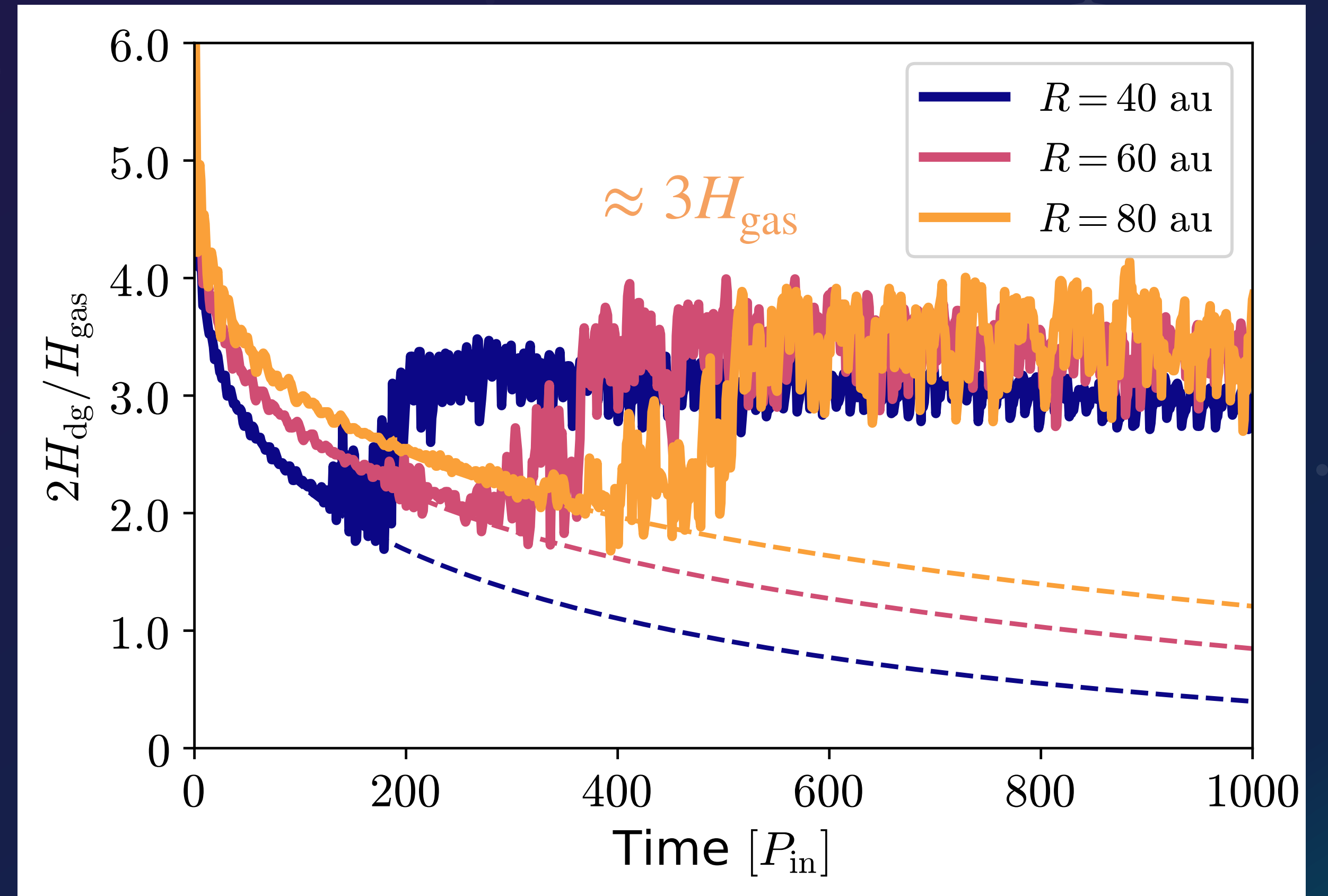
$$2H_{\text{dg}} \approx 3H_{\text{gas}}$$

dashed lines: theoretical prediction without turbulence, following the equation of

$$\frac{d^2 z_{\text{dust}}}{dt^2} = -\Omega_K^2 z_{\text{dust}} - \frac{1}{t_{\text{stop}}} \frac{dz_{\text{dust}}}{dt}$$

with $z_{\text{dust}} = \max\{z\}$ and $dz_{\text{dust}}/dt = 0$ at $t = 0$

time evolution of dust layer's thickness



Turbulence Diagnostics

- ❖ From dust layer's thickness, we estimate **dust vertical diffusion coefficient** α_{diff}

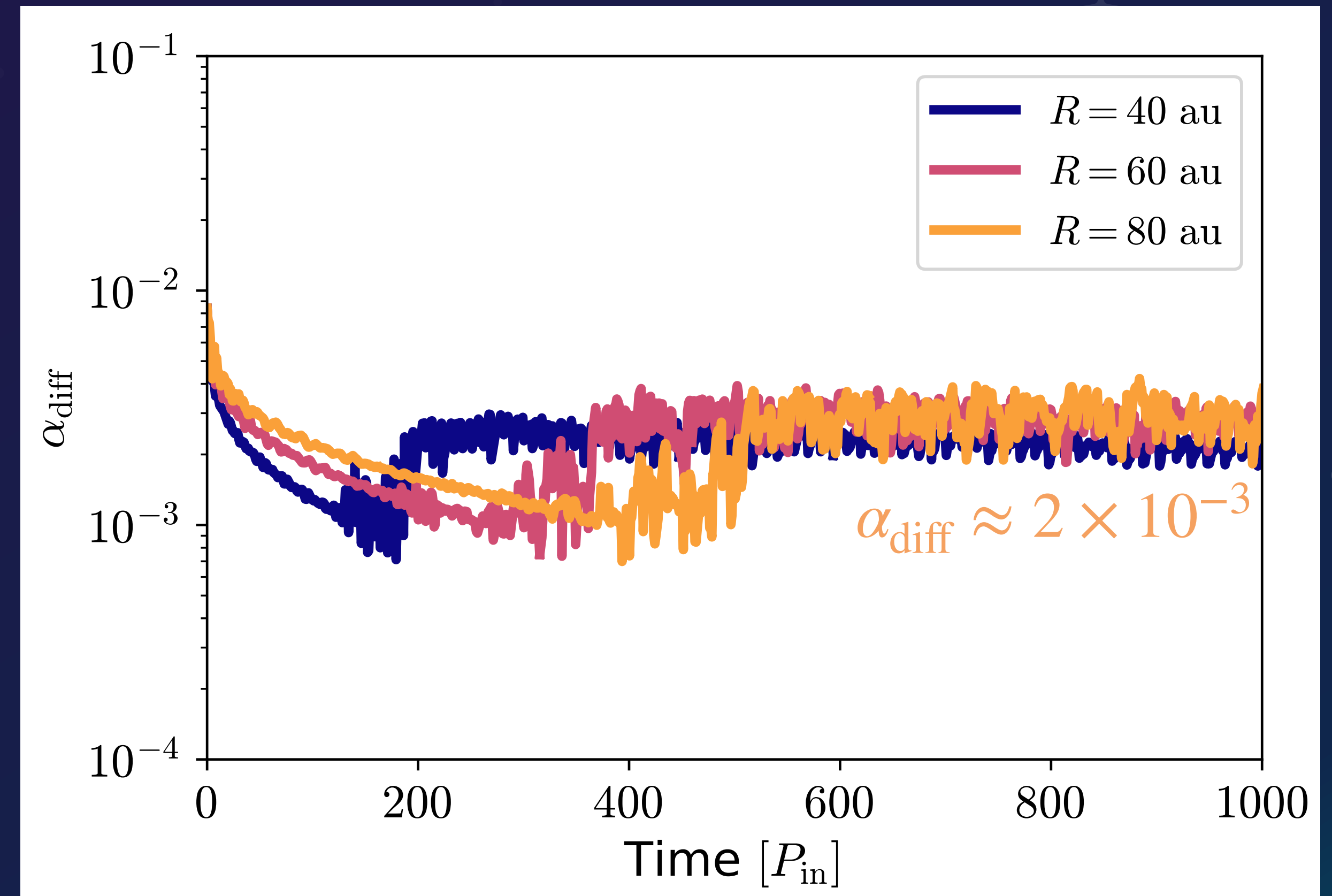
$$H_{\text{dg}} = \left(\frac{\text{St}}{\alpha_{\text{diff}}} \right)^{-1/2} H_{\text{gas}} \quad \text{Dubrulle et al. 1995}$$

- St is the Stokes number $\text{St} = t_{\text{stop}} \Omega_{\text{K}} \sim 10^{-4}$

- ❖ In equilibrium state, α_{diff} converges to

$$\alpha_{\text{diff}} \approx 2 \times 10^{-3}$$

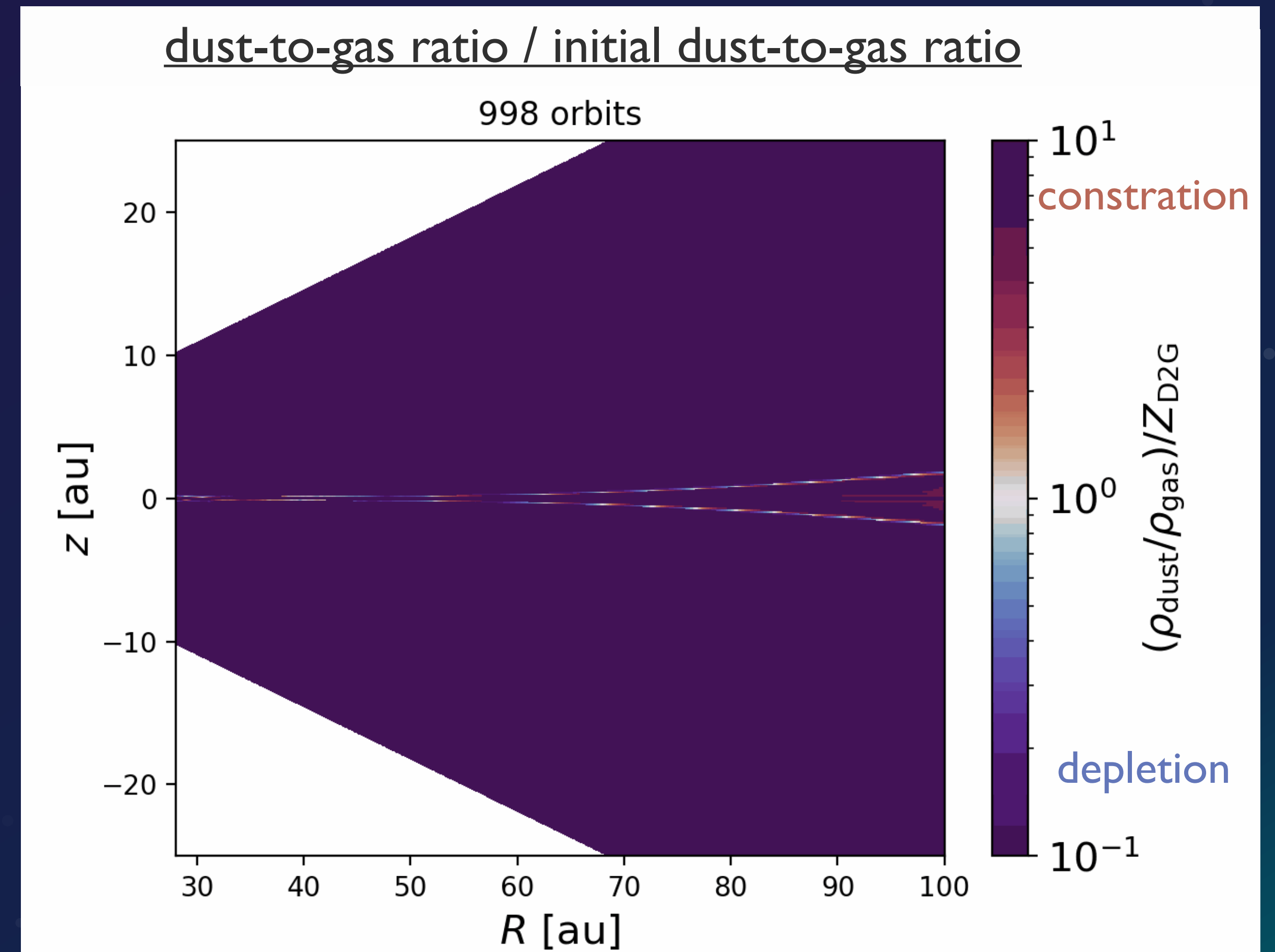
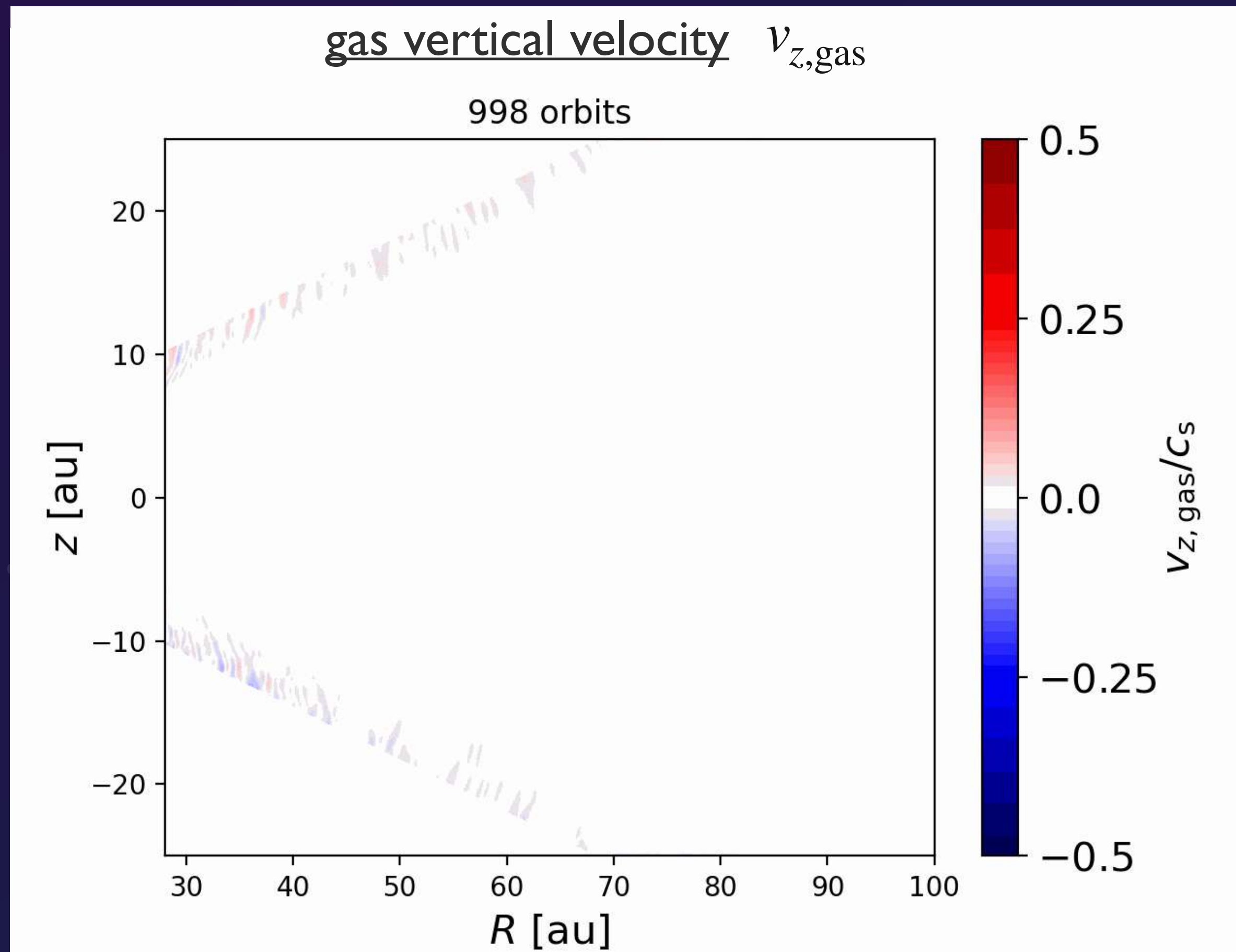
time evolution of dimensional vertical diffusion coefficient



No turbulence

- ❖ In cases of **larger grain size**, VSI cannot drive turbulence
- ❖ Dust continues to settle toward midplane because VSI cannot sustain dust diffusion

grain size = $100 \mu\text{m}$
dust-to-gas ratio = 0.01



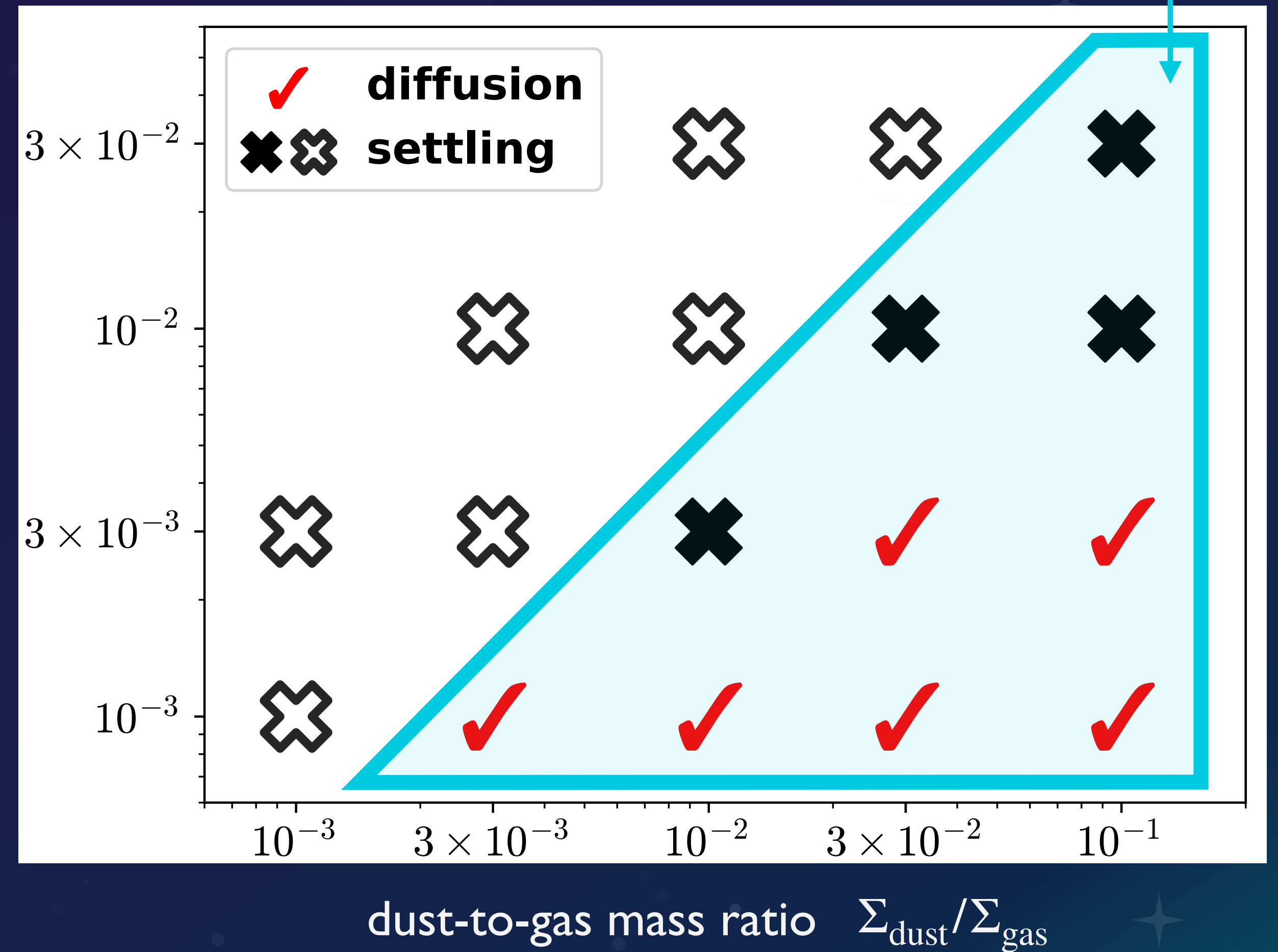
Diffusion vs. Settling

- ❖ Equilibrium state can exist only for **small dust** or **high dust-to-gas mass ratio**
- ❖ For large dust and low dust-to-gas mass ratio, dust settles toward midplane
- ❖ Even if VSI-active layers exist initially, VSI cannot sustain dust diffusion for large dust

grain size a [cm]

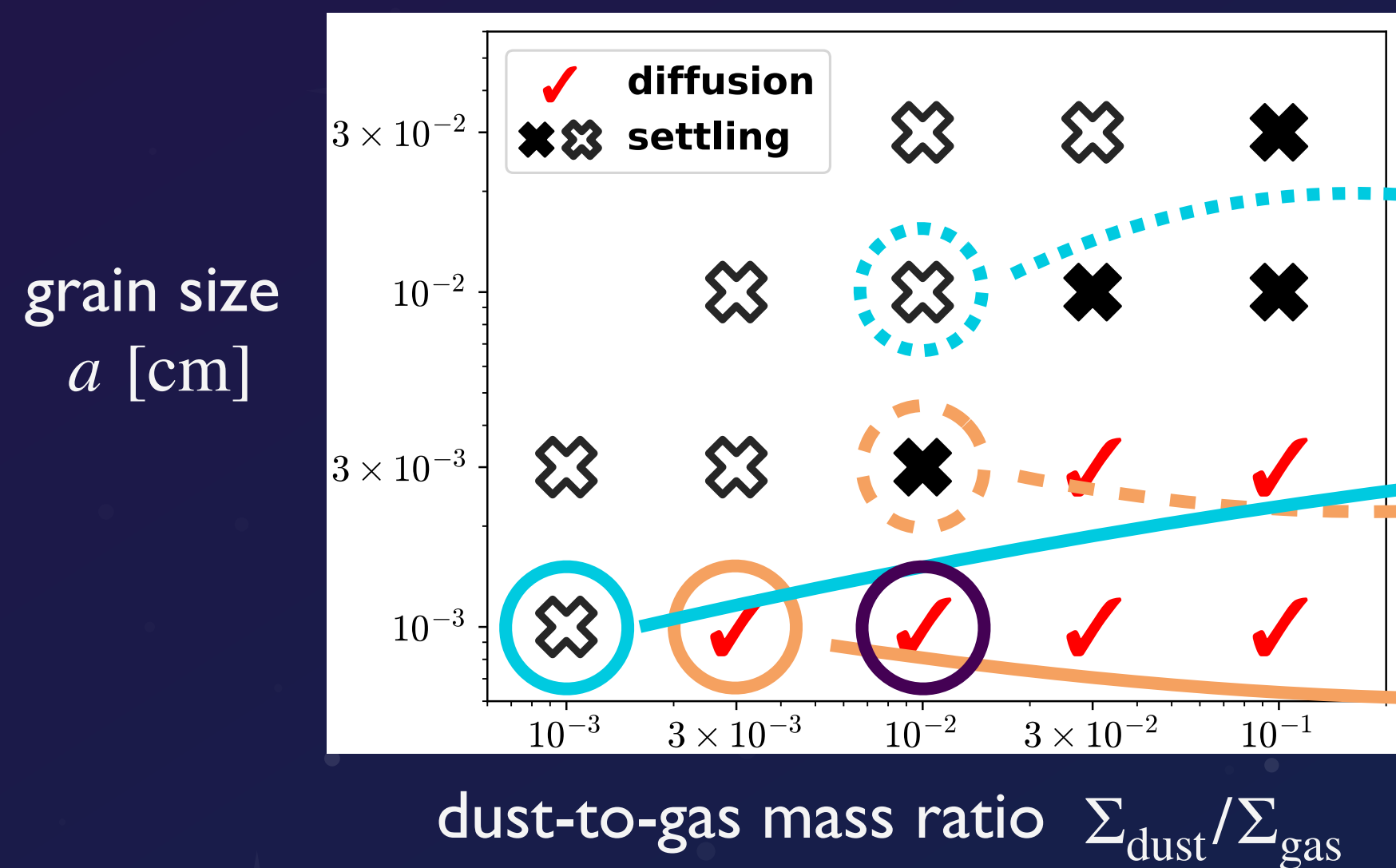
summary of simulations

VSI-active region is present at the initial condition

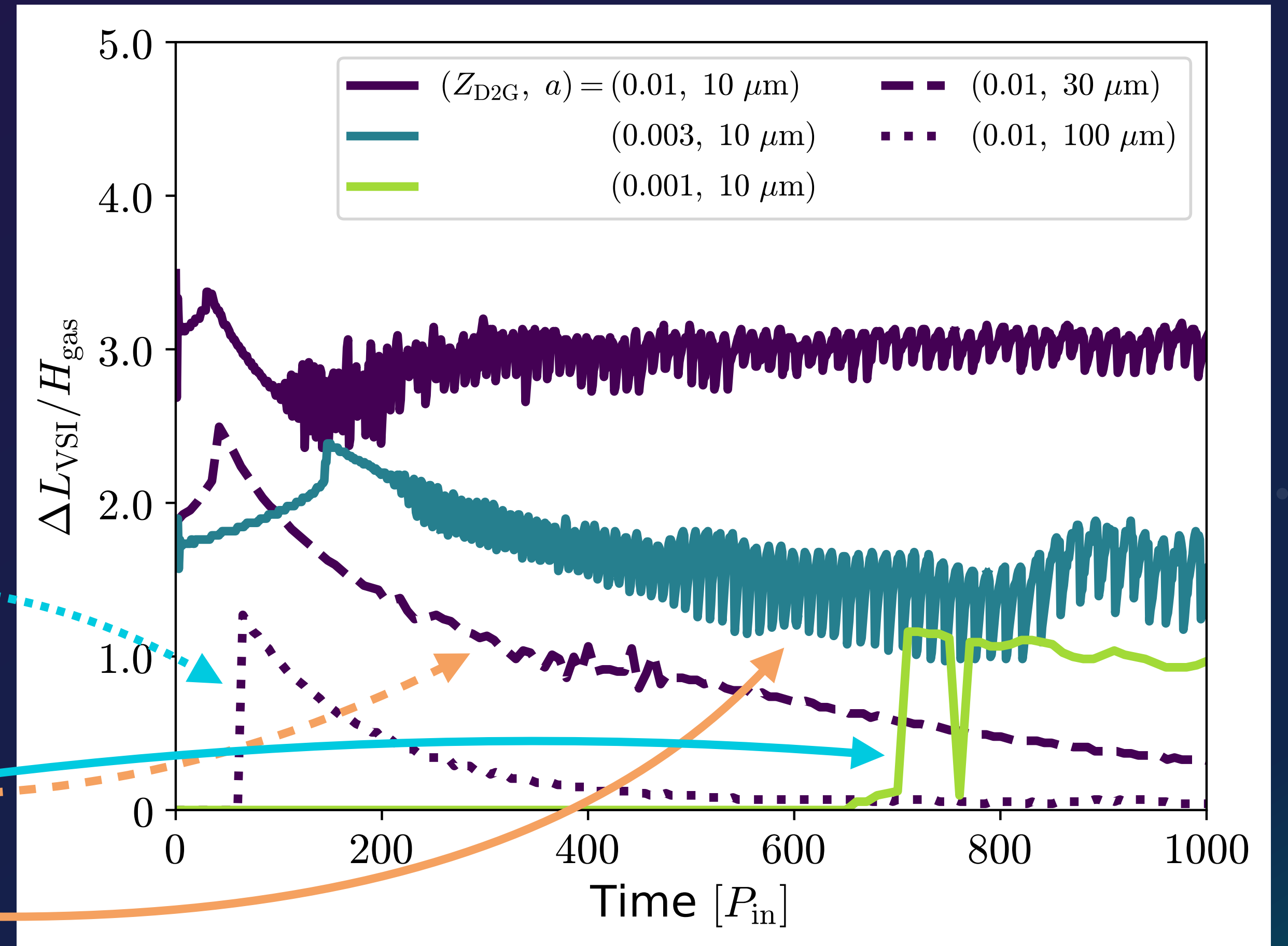


Existence of equilibrium state

- ❖ Even if VSI-active layers exist at initial condition, final state is different
- ❖ Even if no VSI-active layer is present initially, it forms as dust settles.
 - But, this layer is too thin to generate turbulence



time evolution of VSI-active layer's thickness



Simulations as shown previous slides

❖ Our simulations have started from an initial state with no turbulence

- ▶ no turbulence
(random perturbation)
- ▶ uniform spatial profiles of
dust and cooling time



equilibrium state for
VIS turbulence and dust distribution

What would happen if VSI had been driving turbulence from the start?

- ▶ **saturated VSI turbulence**
- ▶ uniform spatial profiles of
dust and cooling time



?

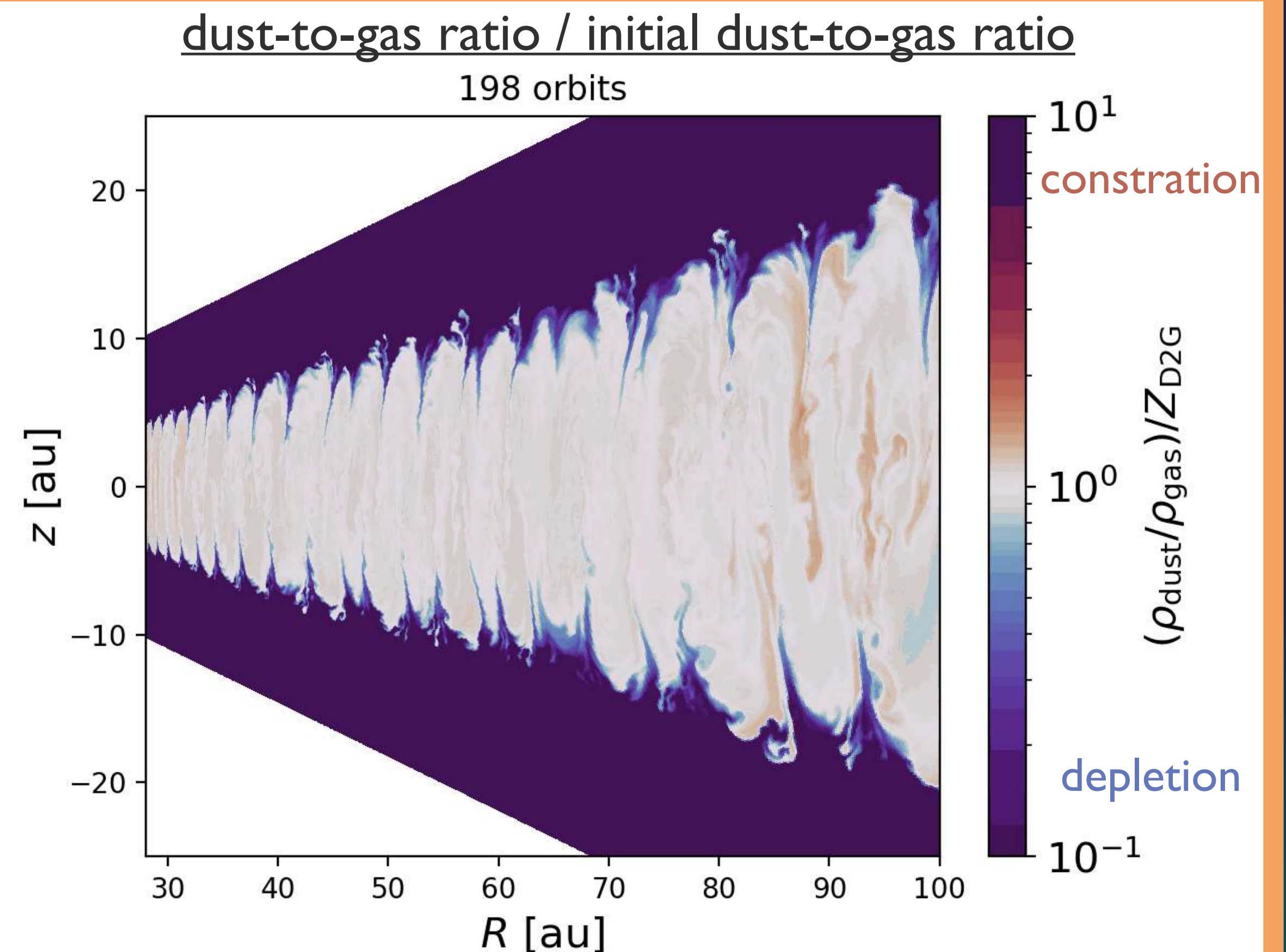
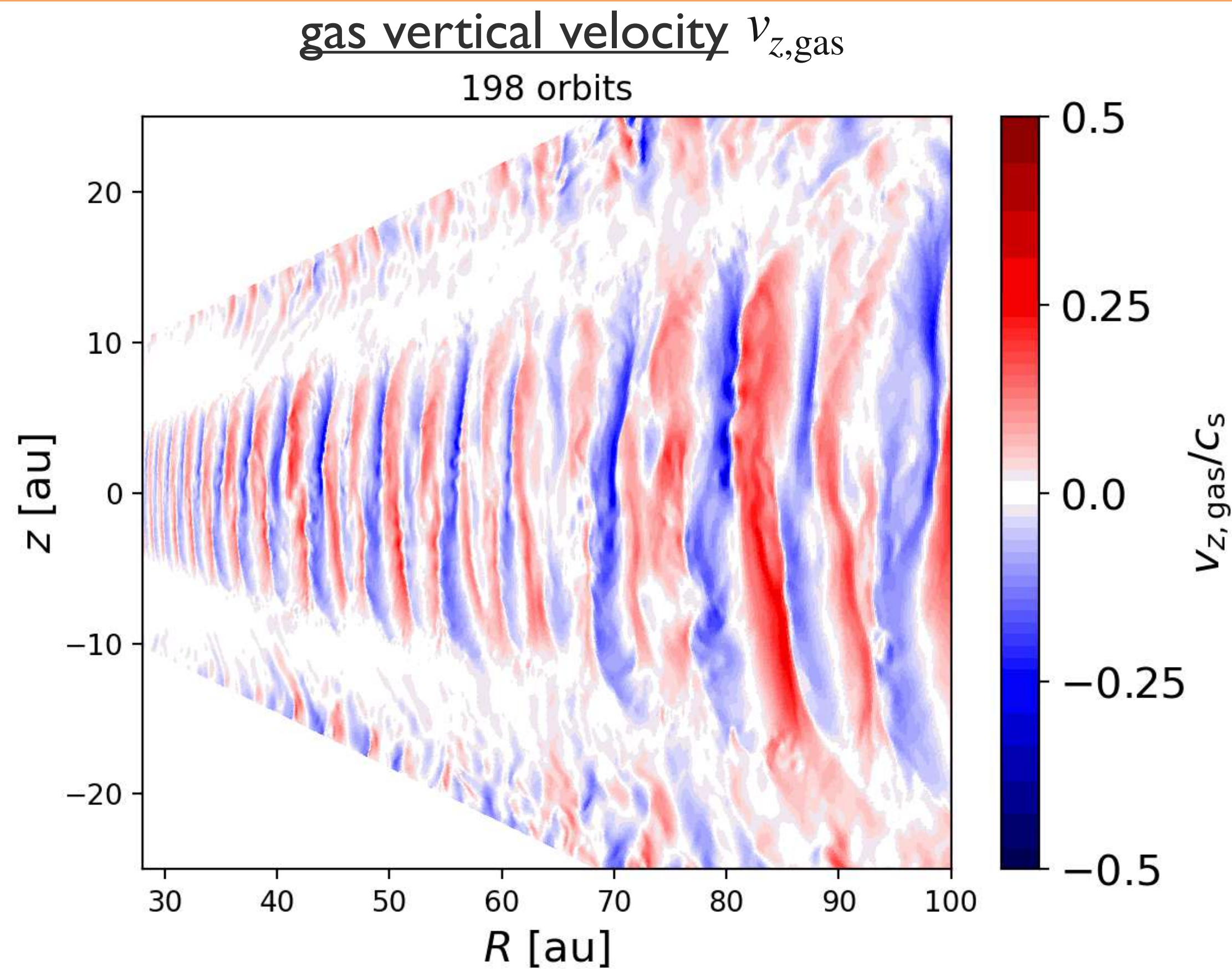
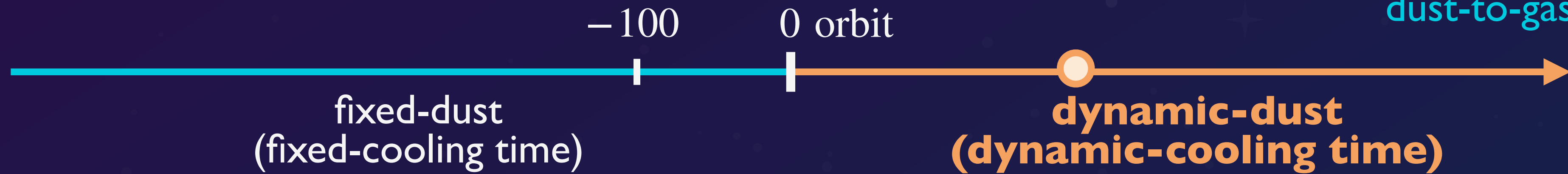
Independence of initial condition

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- ❖ This equilibrium does not depend on the initial condition

grain size = $10\ \mu\text{m}$

dust-to-gas ratio = 0.01



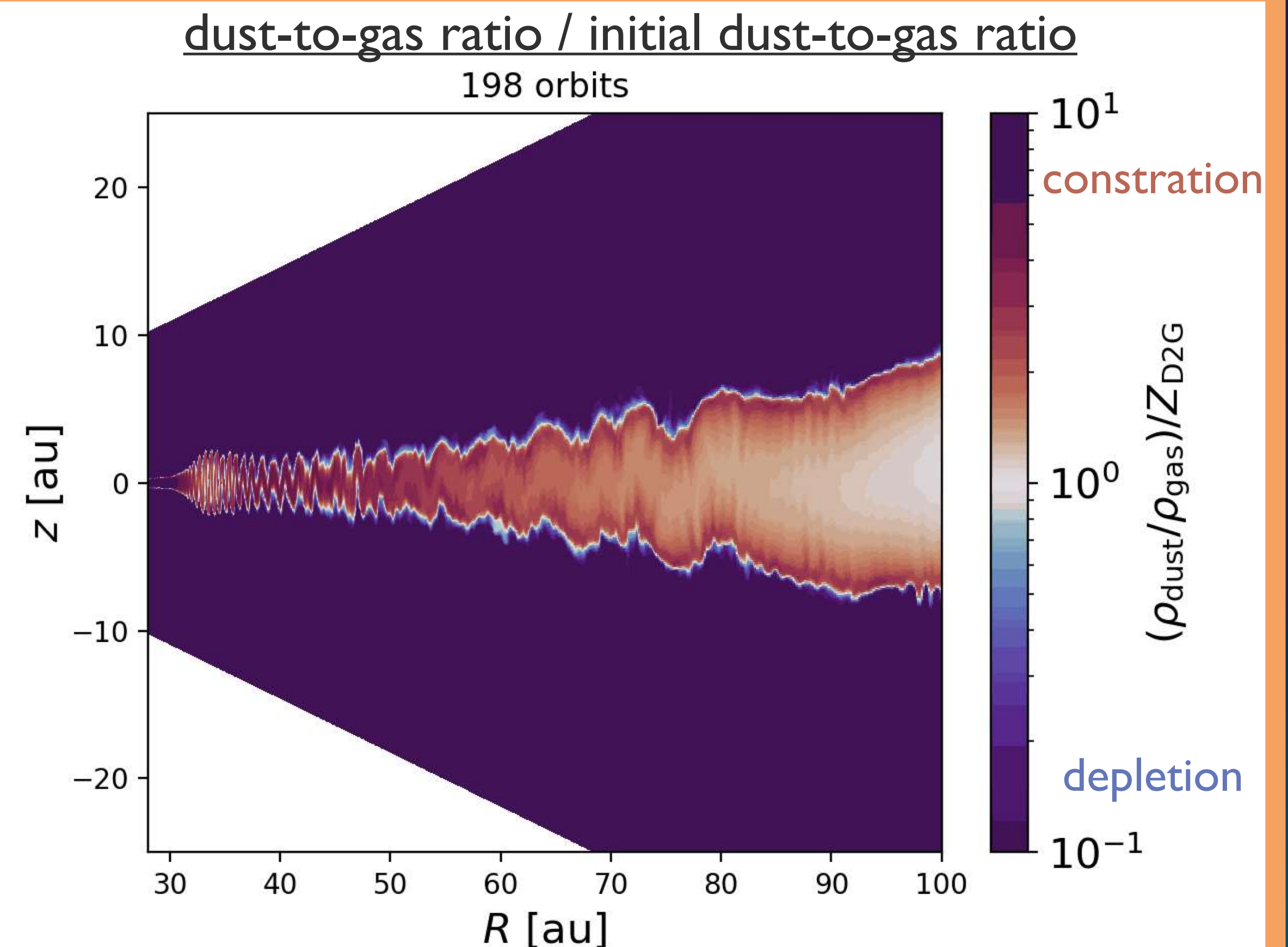
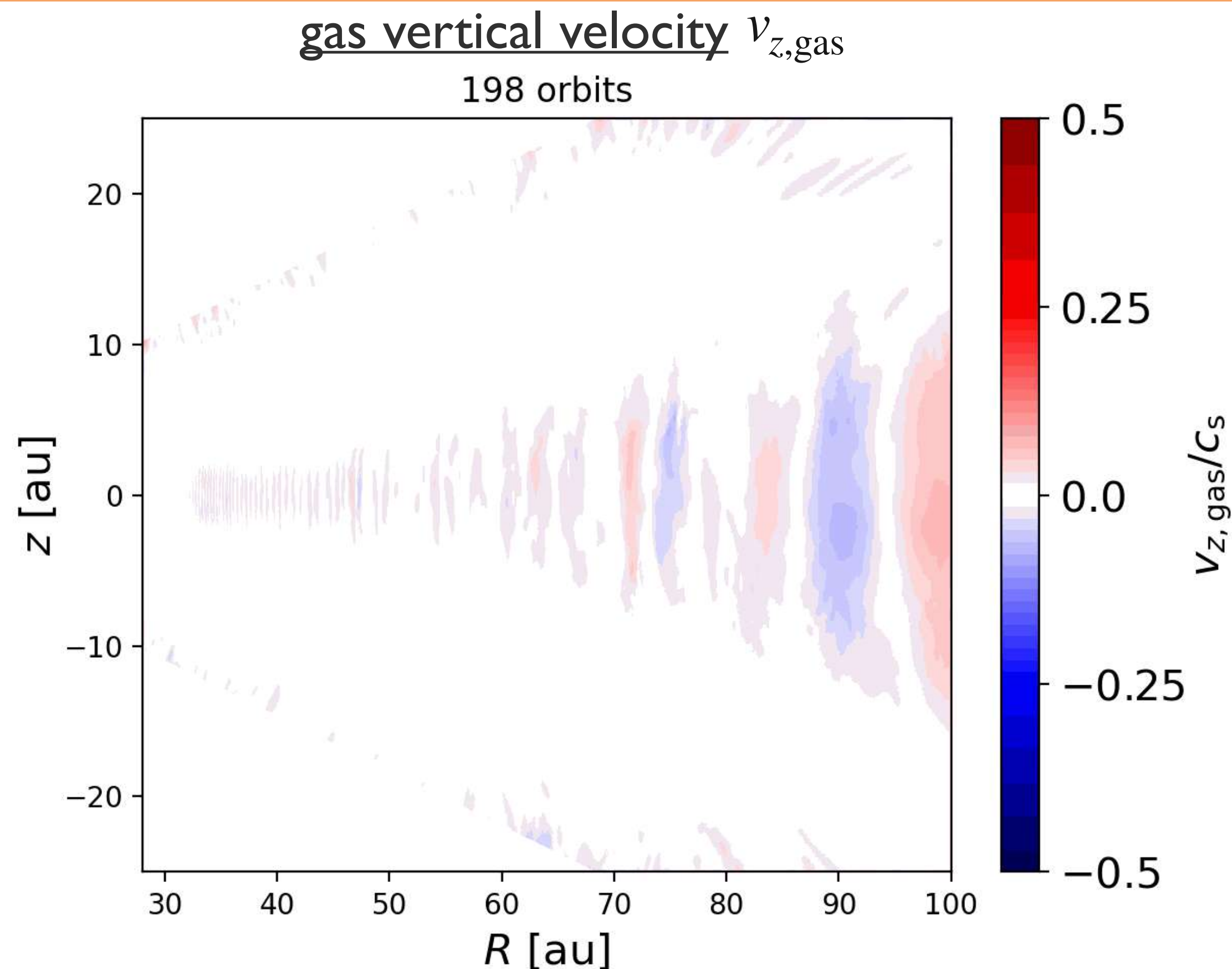
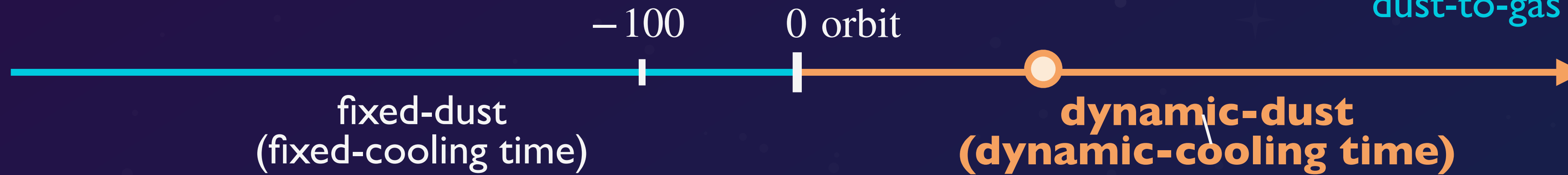
Dust settling while suppressing turbulence

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- ❖ Even if VSI turbulence operates initially, turbulence is suppressed due to dust settling

grain size = $100\ \mu\text{m}$

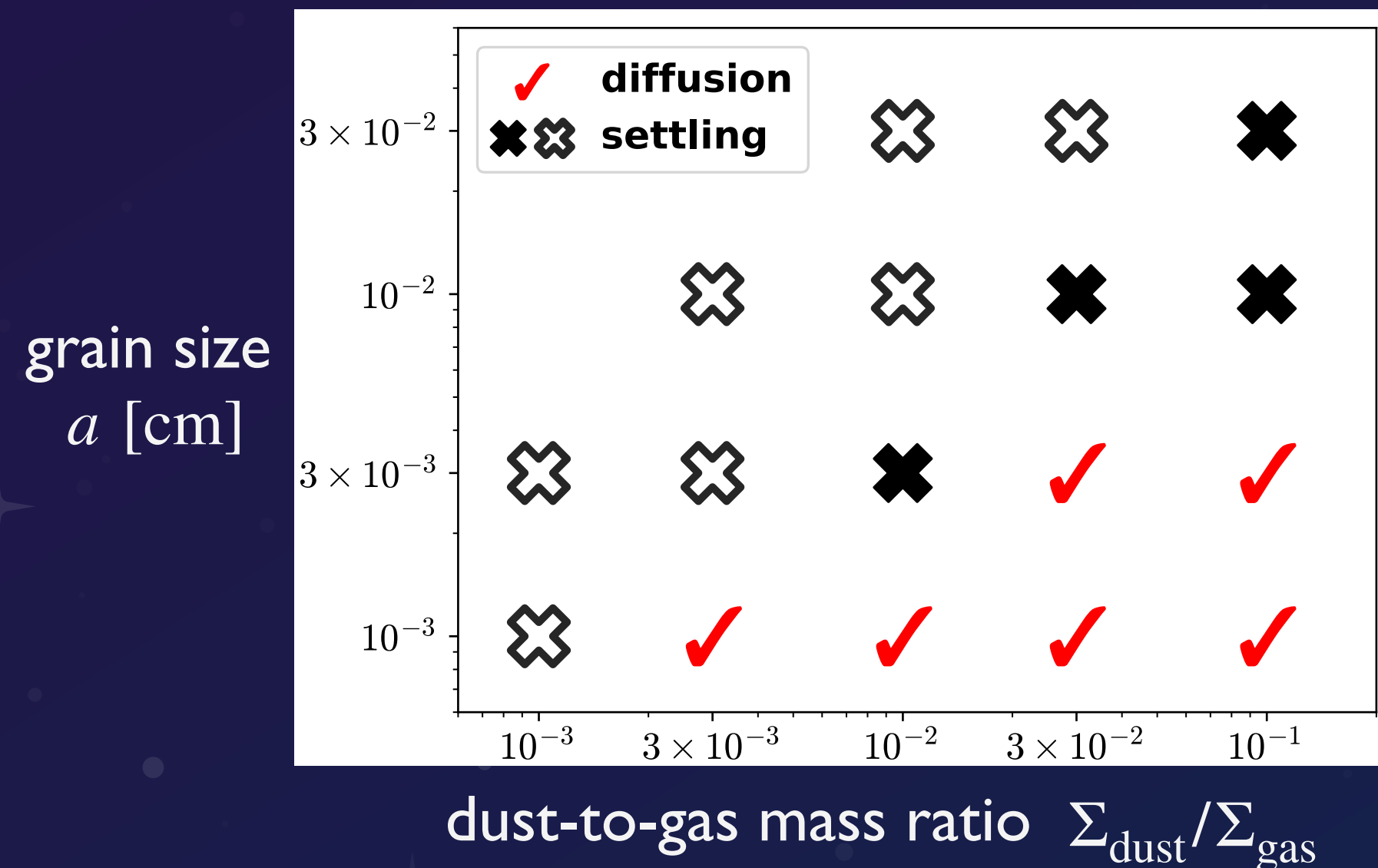
dust-to-gas ratio = 0.03



Diffusion vs. Settling

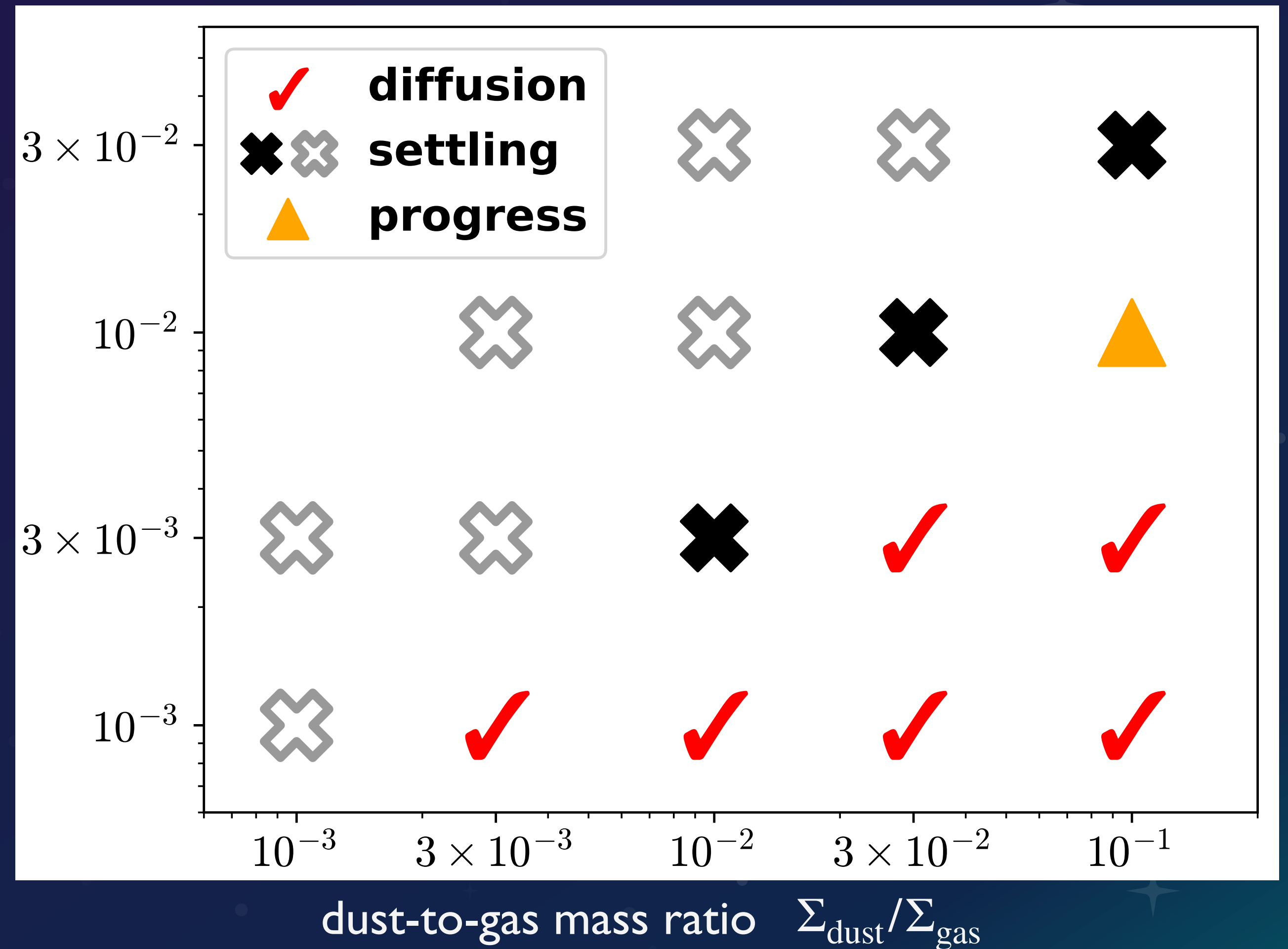
- ❖ Condition for equilibrium state hold no matter whether or not VSI turbulence operates initially
 - ▶ One run is for a state in which equilibrium has not been reached and turbulence is not completely eliminates

initially random perturbation



grain size
 a [cm]

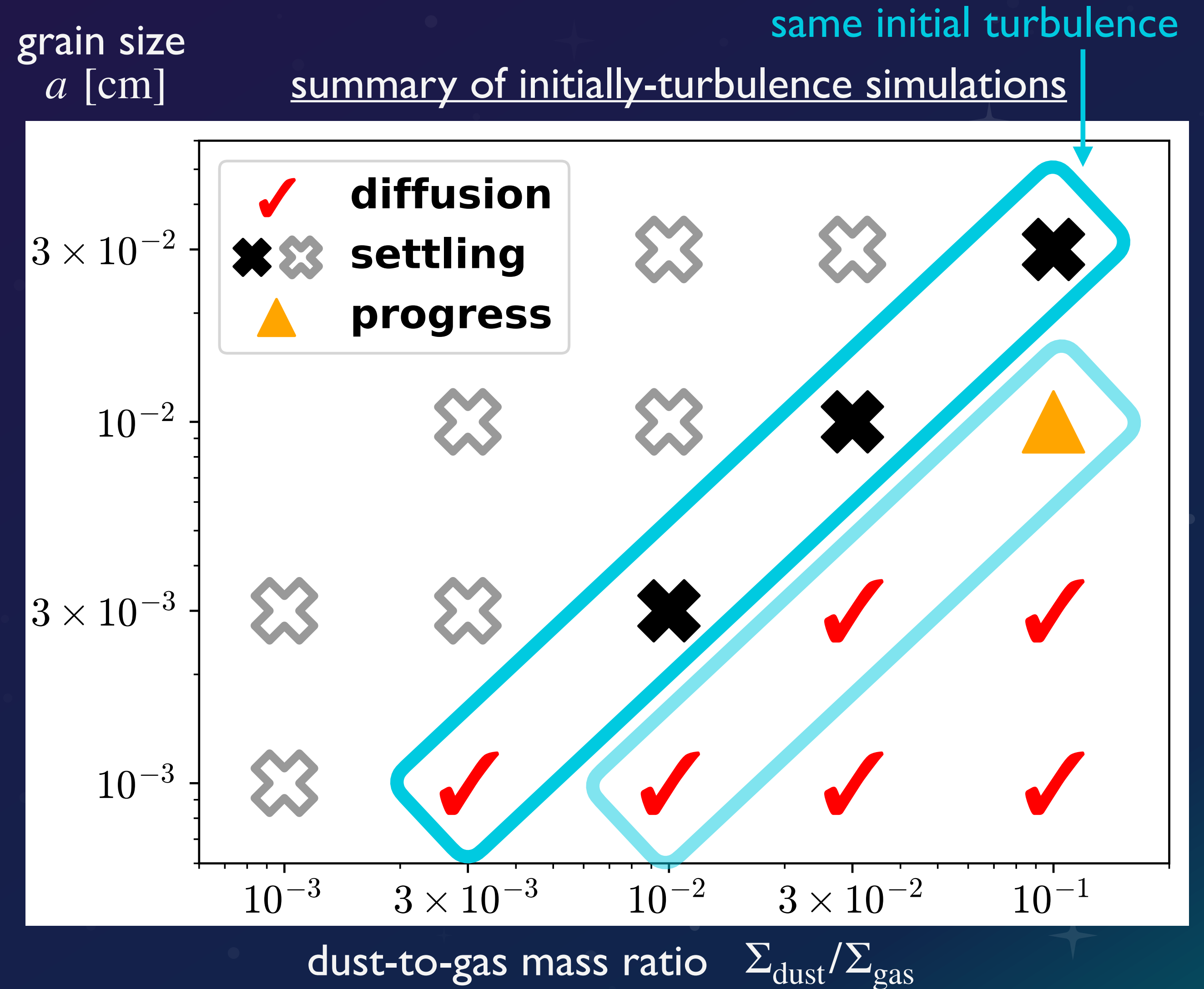
summary of initially-turbulence simulations



Our results

- ❖ Initial vertical profile of cooling timescale cannot determine existence of an equilibrium state
- ❖ Same initial vertical distribution of cooling timescale, but the final state is different

What makes different?

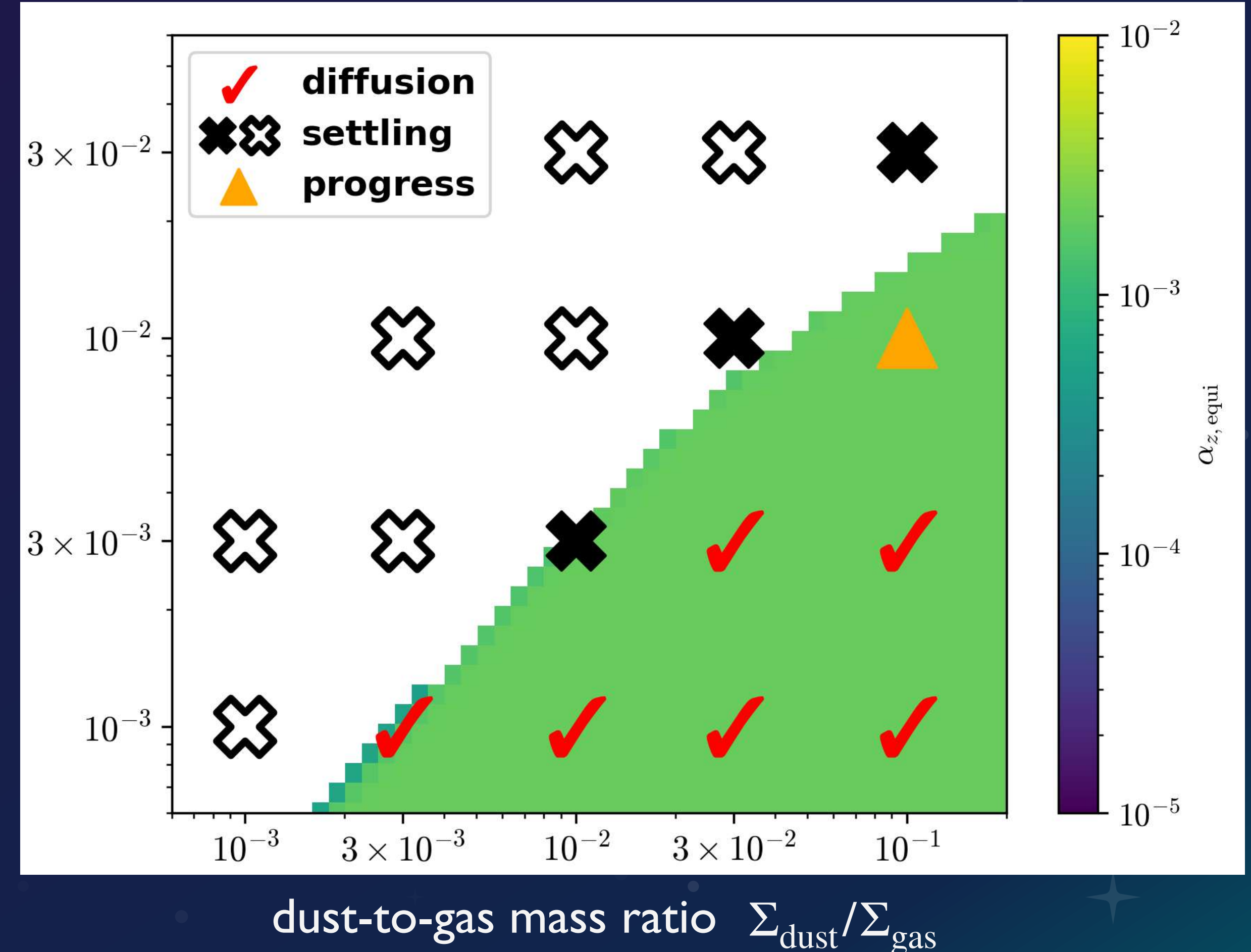


Comparison with semi-analytic model

grain size
 a [cm]

summary & analytical equilibrium diffusion coefficient

- ❖ We calculate equilibrium vertical diffusion coefficient, using semi-analytic model of *Fukuhara & Okuzumi 2024*
 - ▶ Self-consistent model that determines vertical dust distribution and VSI turbulence intensity
- ❖ The model can predict the equilibrium state!
- ❖ Balance between vertical turbulent diffusion and settling determines whether equilibrium states can be achieved

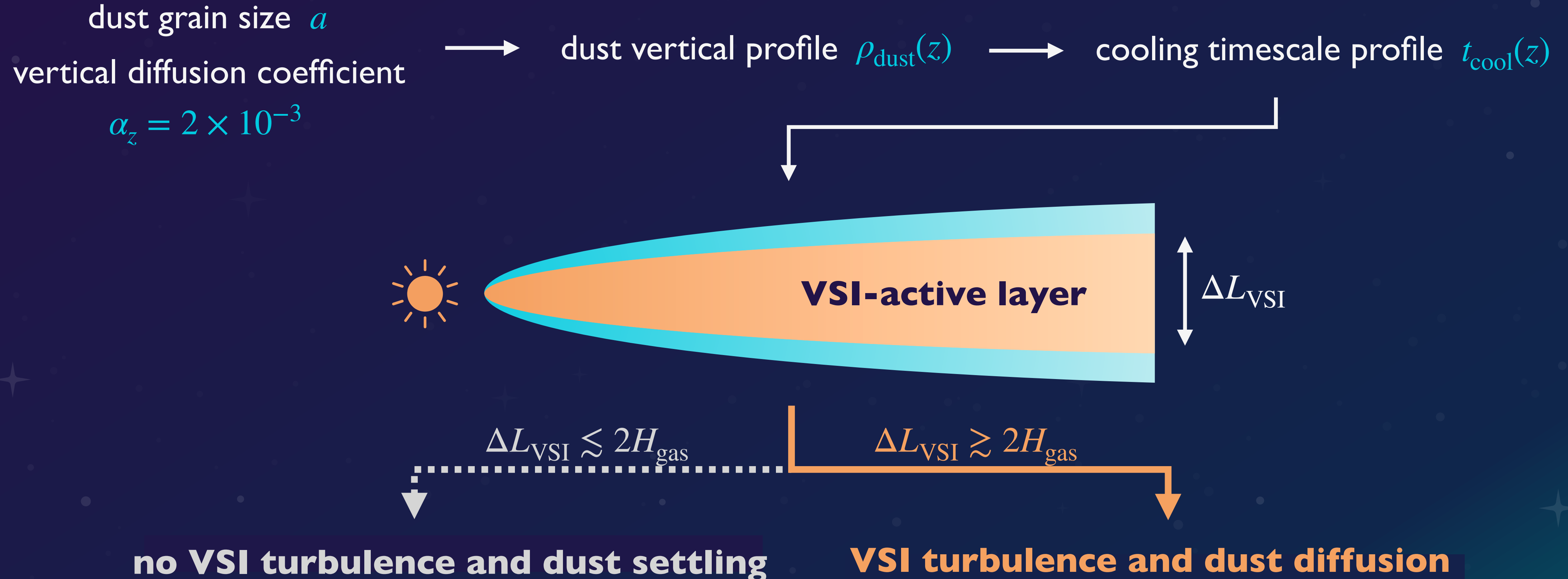


Cooling layer's thickness determines the onset of VSI turbulence

- ❖ Assuming vertical profiles of dust and cooling timescale with $\alpha_z = 2 \times 10^{-3}$, condition under which VSI drives turbulence is

$$\Delta L_{\text{VSI}} \gtrsim 2H_{\text{gas}}$$

Flowchart



Fukuhara, Flock, Okuzumi, Tominaga, A&A, 701, A72 (2025)

- ❖ There exist the equilibrium state for VSI turbulence and dust vertical profile when dust grains are small
- ❖ For large dust grains, VSI cannot drive turbulence due to ineffective cooling
- ❖ Condition for equilibrium state hold no matter whether or not VSI turbulence operates initially
- ❖ Whether equilibrium is achieved can depend on balance between vertical turbulent diffusion and settling
- ❖ **VSI needs a large amount of small dust!**